

Mathematics for Aerospace Engineers

Chapter 5: Vector Calculus

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5 Vector Calculus

5.1 Introduction

We are familiar with the way that single variable differential calculus can be extended to consider functions of two or more variables and differentials of scalar quantities expressed in terms of partial derivatives. An example might be temperature $T = T(x, y, z, t)$ as a scalar varying with position (x, y, z) and time t .

We can carry these ideas over to consider vector functions, where the components of the vector can vary with spatial coordinates or time. Examples include force $\mathbf{F}$ or velocity $\mathbf{v}$, each generally with three spatial components.

To analyse the differential properties of vectors, we need to introduce new differential operators, which also have distinct interpretations.

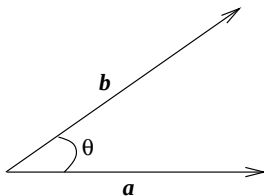
5.2 Reminder of some vector properties

The *dot product* or *scalar product* of two vectors is a scalar quantity,

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3 = |\mathbf{a}| |\mathbf{b}| \cos \theta,$$

where θ is the angle between the vectors:

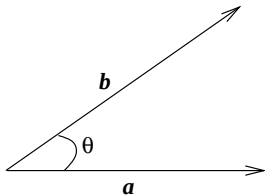
Note: $\hat{\mathbf{a}} \cdot \hat{\mathbf{b}} = \cos \theta$, where $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$ are unit vectors in the direction of $\mathbf{a}$ and $\mathbf{b}$, respectively.



Properties of the scalar product

- $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$.
- If $\mathbf{a}$ and $\mathbf{b}$ are perpendicular then $\mathbf{a} \cdot \mathbf{b} = 0$.
- $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$.

The *cross product* or *vector product* of two vectors $\mathbf{a} \times \mathbf{b}$ is a vector quantity, with magnitude $|\mathbf{a}||\mathbf{b}| \sin \theta$ and direction perpendicular to both $\mathbf{a}$ and $\mathbf{b}$ in a right-handed sense:



The direction of $\mathbf{a} \times \mathbf{b}$ is out of the page.

In components,

$$\begin{aligned}\mathbf{a} \times \mathbf{b} &= (a_2b_3 - a_3b_2)\mathbf{i} + (a_3b_1 - a_1b_3)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k} \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}.\end{aligned}$$

Properties of the vector product

- $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}.$
- If $\mathbf{a}$ and $\mathbf{b}$ are parallel then, $\mathbf{a} \times \mathbf{b} = \mathbf{0}.$
- $\mathbf{a} \times \mathbf{a} = \mathbf{0}.$

5.2.1 Derivatives of vector valued functions

If a vector $\mathbf{h}$ varies with respect to a scalar variable u then

$$\mathbf{h}(u) = h_1(u)\mathbf{i} + h_2(u)\mathbf{j} + h_3(u)\mathbf{k},$$

and if h_1 , h_2 and h_3 are differentiable

$$\mathbf{h}'(u) = h'_1(u)\mathbf{i} + h'_2(u)\mathbf{j} + h'_3(u)\mathbf{k},$$

ie a vector valued function is differentiated by differentiating each of the components.

We may alternatively write

$$\mathbf{h}(u) = (h_1(u), h_2(u), h_3(u))$$

and then

$$\mathbf{h}'(u) = (h'_1(u), h'_2(u), h'_3(u)).$$

5.2.1 Derivatives of vector valued functions (ctd)

If $\alpha = \alpha(u)$ is a differentiable scalar function and $\mathbf{h} = \mathbf{h}(u)$ and $\mathbf{g} = \mathbf{g}(u)$ are differentiable vector functions then the following rules hold.

Differentiation rules for vector valued functions

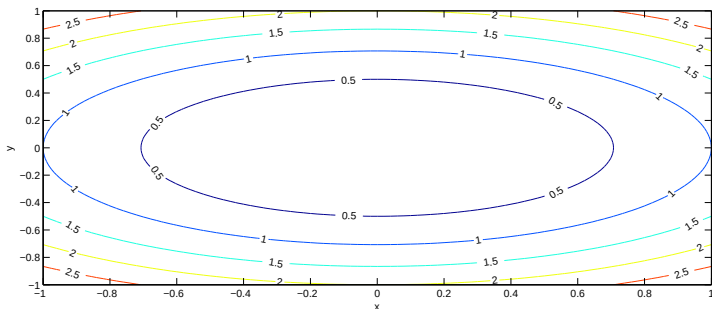
- $\frac{d}{du}(\alpha \mathbf{h}) = \frac{d\alpha}{du} \mathbf{h} + \alpha \frac{d\mathbf{h}}{du}$
- $\frac{d}{du}(\mathbf{h} \cdot \mathbf{g}) = \mathbf{h} \cdot \frac{d\mathbf{g}}{du} + \frac{d\mathbf{h}}{du} \cdot \mathbf{g}$
- $\frac{d}{du}(\mathbf{h} \times \mathbf{g}) = \mathbf{h} \times \frac{d\mathbf{g}}{du} + \frac{d\mathbf{h}}{du} \times \mathbf{g}$

5.3 Scalar and vector fields

A scalar or vector quantity is said to be a *field* if it is a function of position. For example,

- The temperature in a room is defined $T(x, y, z) = T(\mathbf{r})$ and is a scalar field

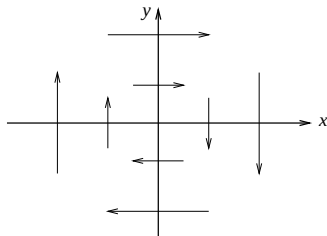
Example $T(x, y) = x^2 + 2y^2$ is a scalar field. Some geometrical features are given by the contour lines $T(x, y) = \text{const}$ (ellipses).



- The wind velocity depends on where in space you are, ie $\mathbf{v}(x, y, z) = \mathbf{v}(\mathbf{r})$, which is a vector field.

Example 5.3.1

If $\mathbf{v}(x, y) = (y, -x)$, the plot of $\mathbf{v}$, called the field plot, shows some geometrical features of the field.



The length of the arrows indicates $|\mathbf{v}|$ at each point.

5.4 Vector differential operators

The vector differential operator ∇ (called 'del' or 'nabla') is given by

$$\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$$

(in a cartesian coordinate system), and can act (similar to a vector) on scalar or vector functions.

Let's look at how it acts on a scalar field first.

5.5 Gradient of a scalar field

Suppose $f(x, y, z) = f(\mathbf{r})$ is a scalar field. Then ∇ transforms it into a vector field, called the *gradient* of f .

$$\text{grad } f = \nabla f = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k}$$

Example 5.5.1

If $f(x, y, z) = x^2 + y^2 + z$, then

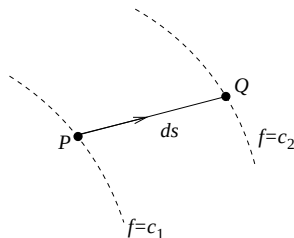
$$\frac{\partial f}{\partial x} = 2x, \quad \frac{\partial f}{\partial y} = 2y, \quad \frac{\partial f}{\partial z} = 1$$

and

$$\nabla f = \begin{pmatrix} 2x \\ 2y \\ 1 \end{pmatrix}.$$

5.5.1 The physical meaning of the gradient

Consider two points $P = (x, y, z)$ and $Q = (x + dx, y + dy, z + dz)$ with “infinitesimal” dx , dy and dz , and a scalar field f .

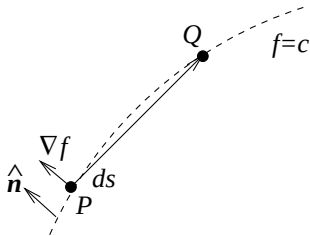


Then $\overrightarrow{PQ} = \mathbf{ds} = (dx, dy, dz)$ and (from the first-order Taylor expansion of f) the change of f is approximately

$$\begin{aligned} df = f(Q) - f(P) &\approx \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz \\ &= \nabla f \cdot \mathbf{ds} \end{aligned}$$

The term “infinitesimal” means we may neglect dx^2 , $dx dy$ etc in the Taylor expansion of f , as the sizes of dx , dy and dz approach 0.

If P and Q are on the same level surface (isosurface) then $df = 0$ so in the above $\nabla f \cdot d\mathbf{s} \approx 0$.



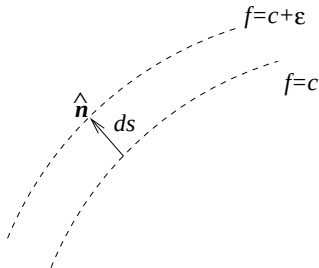
Hence ∇f and $d\mathbf{s}$ are approximately perpendicular for small $d\mathbf{s}$. For infinitesimal $d\mathbf{s}$ (ie infinitesimally small distance between P and Q), the vector $d\mathbf{s}$ is a tangent vector to the level surface, that is ∇f is perpendicular to the surfaces $f = \text{const}$. Therefore $\nabla f = |\nabla f| \hat{n}$ where

$$\hat{n} = \frac{\nabla f}{|\nabla f|}.$$

is the unit normal vector to the surface. The vector $\hat{n}$ points in the direction of *fastest increase* of values of f .

If $\mathbf{ds} = dn \hat{\mathbf{n}}$, then

$$\begin{aligned} df &= \nabla f \cdot \mathbf{ds} = \nabla f \cdot \hat{\mathbf{n}} dn = |\nabla f| \hat{\mathbf{n}} \cdot \hat{\mathbf{n}} dn \\ &= |\nabla f| dn. \end{aligned}$$



Hence

$$|\nabla f| = \frac{df}{dn},$$

the rate of change of f in the direction $\hat{\mathbf{n}}$, and $\nabla f = \frac{df}{dn} \hat{\mathbf{n}}$.

Example 5.5.2

Find the unit normal to the surface $x^2 + y^2 - z = 0$ at the point $(1, 1, 2)$.

Define $f(x, y, z) = x^2 + y^2 - z$. Then the surface is $f(x, y, z) = 0$, a level surface of f .

The unit normal $\hat{\mathbf{n}}$ to the isosurface of f is given by

$$\hat{\mathbf{n}} = \frac{\nabla f}{|\nabla f|},$$

where

$$\nabla f = 2x\mathbf{i} + 2y\mathbf{j} - \mathbf{k}, \quad \text{and} \quad |\nabla f| = \sqrt{4x^2 + 4y^2 + 1}.$$

Hence

$$\hat{\mathbf{n}} = \frac{2x\mathbf{i} + 2y\mathbf{j} - \mathbf{k}}{\sqrt{4x^2 + 4y^2 + 1}}.$$

Evaluated at $(1, 1, 2)$ we obtain

$$\hat{\mathbf{n}} = \left(\frac{2}{3}, \frac{2}{3}, -\frac{1}{3} \right).$$

5.5.2 The directional derivative

Take any vector $\mathbf{s}$, and imagine walking in direction $\mathbf{s}$ for a very small amount of time ds . How does f change? We move by displacement

$$d\mathbf{s} = (dx, dy, dz) \text{ and } \mathbf{s} = \left(\frac{dx}{ds}, \frac{dy}{ds}, \frac{dz}{ds} \right)$$

As before (by looking at the Taylor series), f changes by approximately

$$df \approx \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz,$$

and hence

$$\begin{aligned} \frac{df}{ds} &\approx \frac{\partial f}{\partial x} \frac{dx}{ds} + \frac{\partial f}{\partial y} \frac{dy}{ds} + \frac{\partial f}{\partial z} \frac{dz}{ds} \\ &= \nabla f \cdot \mathbf{s} \\ &= |\nabla f| \hat{\mathbf{n}} \cdot \mathbf{s} = |\nabla f| |\mathbf{s}| \cos \theta, \end{aligned}$$

where θ is the angle between $\mathbf{s}$ and $\hat{\mathbf{n}}$.

The above approximation becomes equality as our tiny change in time ds tends to 0.

We thus define the **directional derivative** of f in the direction $\mathbf{s}$ to be

$$\frac{df}{ds} = \nabla f \cdot \mathbf{s}.$$

This quantity gives the ‘instantaneous rate of change of f ’ when one travels in direction $\mathbf{s}$.

If $|\mathbf{s}| = 1$ (note: some texts always assume this restriction to unit vectors) then

$$\frac{df}{ds} = |\nabla f| \cos \theta.$$

where θ is the angle between $\mathbf{s}$ and $\hat{\mathbf{n}} = \frac{\nabla f}{|\nabla f|}$. Then:

- f increases most quickly (at rate $|\nabla f|$) in direction $\mathbf{s} = \hat{\mathbf{n}}$.
- f decreases most quickly (at rate $-|\nabla f|$) in direction $\mathbf{s} = -\hat{\mathbf{n}}$.
- f is stationary in directions orthogonal to $\hat{\mathbf{n}}$, that is, in directions tangent to the isosurface.

Example 5.5.3

Find the maximum rate of change of the scalar field

$$\phi = x^2 + 2yz^2$$

at the point $(2, 1, 1)$, and find the directional derivative in the direction of the vector $\mathbf{p} = (1, 1, 0)$, ie in the direction $\hat{\mathbf{p}} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$.

At the point $(2, 1, 1)$

$$\nabla\phi = (2x, 2z^2, 4yz) = (4, 2, 4),$$

so the maximum rate of change is

$$|\nabla\phi| = \sqrt{4^2 + 2^2 + 4^2} = 6.$$

The rate of change in the direction $\hat{\mathbf{p}} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$ is

$$\frac{1}{\sqrt{2}}(4, 2, 4) \cdot (1, 1, 0) = \frac{6}{\sqrt{2}}.$$

5.5.3 Summary

Gradients and directional derivatives

If $f(x, y, z) = f(\mathbf{r})$ is a differentiable scalar field then

- $\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k}$
- $\frac{df}{ds} = \nabla f \cdot \mathbf{s}$ for any vector $\mathbf{s}$
- $\nabla f = \frac{df}{dn} \hat{\mathbf{n}} = |\nabla f| \hat{\mathbf{n}}$, where $\hat{\mathbf{n}}$ is the unit normal vector to the level surface $f = \text{const.}$ Hence
- $\hat{\mathbf{n}} = \frac{\nabla f}{|\nabla f|}$

5.6 Divergence of a vector field

Given a vector field $\mathbf{A}(\mathbf{r})$ with components

$$\mathbf{A}(\mathbf{r}) = (A_1(\mathbf{r}), A_2(\mathbf{r}), A_3(\mathbf{r})),$$

the divergence of $\mathbf{A}$ is a scalar field, defined as

$$\operatorname{div} \mathbf{A} = \frac{\partial A_1}{\partial x} + \frac{\partial A_2}{\partial y} + \frac{\partial A_3}{\partial z}.$$

This can be written in terms of the symbol ∇ ,

$$\operatorname{div} \mathbf{A} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \cdot (A_1, A_2, A_3) = \nabla \cdot \mathbf{A}.$$

Physical interpretation of the divergence

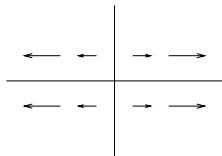
Imagine $\mathbf{A}$ as giving the directions of a flow in 3-space (or in 2d think a weather forecast showing wind directions on a map). Consider how a dense number of particles move around under this flow.

Then $\nabla \cdot \mathbf{A}$ is large (and positive) at $\mathbf{r}$ if particles in a small local vicinity of $\mathbf{r}$ tend to quickly leave the region rather than enter it. Another way to think about this is that, if the pressure were to remain the same, particles would need to be created near to $\mathbf{r}$, making it a 'source'.

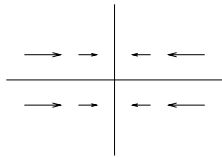
If $\nabla \cdot \mathbf{A}$ is negative at $\mathbf{r}$, then the particles tend to flow *inwards* in a small region about $\mathbf{r}$. One may imagine a 'sink' in that location.

A vector field with $\nabla \cdot \mathbf{A} = 0$ at all locations is sometimes called **incompressible** or **solenoidal**. In many physical situations vector fields will be close to incompressible, namely in applications where one expects the changes in pressure between regions to be negligible.

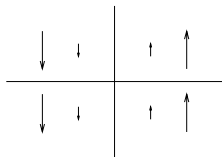
(i) If $\mathbf{u} = (x, 0, 0)$, $\nabla \cdot \mathbf{u} = \frac{\partial x}{\partial x} = 1$ for all $\mathbf{r}$.



(ii) If $\mathbf{u} = (-x, 0, 0)$, $\nabla \cdot \mathbf{u} = \frac{\partial(-x)}{\partial x} = -1$ for all $\mathbf{r}$.



(iii) If $\mathbf{u} = (0, x, 0)$, $\nabla \cdot \mathbf{u} = \frac{\partial x}{\partial y} = 0$ for all $\mathbf{r}$.



Example 5.6.1

If $\mathbf{A} = (2xy, xy^2z, \ln z)$, with $z > 0$, find $\nabla \cdot \mathbf{A}$.

$$\frac{\partial A_1}{\partial x} = 2y, \quad \frac{\partial A_2}{\partial y} = 2xyz, \quad \frac{\partial A_3}{\partial z} = \frac{1}{z}.$$

Hence

$$\nabla \cdot \mathbf{A} = 2y + 2xyz + \frac{1}{z}.$$

Example 5.6.1

If $\mathbf{r} = (x, y, z)$ find $\nabla \cdot \mathbf{r}$.

$$\frac{\partial x}{\partial x} = 1, \quad \frac{\partial y}{\partial y} = 1, \quad \frac{\partial z}{\partial z} = 1.$$

Hence

$$\nabla \cdot \mathbf{r} = 1 + 1 + 1 = 3.$$

5.7 Curl of a vector field

The *curl* of a vector field $\mathbf{A}(\mathbf{r})$ is a vector field, defined by

$$\text{curl } \mathbf{A} = \left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z}, \frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x}, \frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right).$$

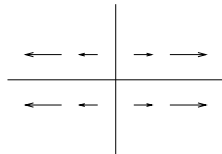
This can be written in the notation of,

$$\text{curl } \mathbf{A} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_1 & A_2 & A_3 \end{vmatrix}.$$

In terms of the symbol ∇ ,

$$\text{curl } \mathbf{u} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \times (A_1, A_2, A_3) = \nabla \times \mathbf{A}.$$

Example 5.7.1



(i) If $\mathbf{u} = (x, 0, 0)$

$$\nabla \times \mathbf{u} = (0, 0, 0)$$

(ii) If $\mathbf{u} = \mathbf{r} = (x, y, z)$

The x -component of $\nabla \times \mathbf{r}$ is

$$\frac{\partial z}{\partial y} - \frac{\partial y}{\partial z} = 0$$

because y and z are independent. Similarly the other components are zero so

$$\nabla \times \mathbf{r} = \mathbf{0}.$$

Example 5.7.2

If

$$\mathbf{A} = (2xy, xy^2z, \ln z),$$

find $\nabla \times \mathbf{A}$.

$$\begin{aligned}\nabla \times \mathbf{A} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy & xy^2z & \ln z \end{vmatrix} = \mathbf{i}(-xy^2) + \mathbf{j}(0) + \mathbf{k}(y^2z - 2x) \\ &= -xy^2\mathbf{i} + (y^2z - 2x)\mathbf{k}\end{aligned}$$

Physically, $\nabla \times \mathbf{u}$ represents the 'rotation' or 'twisting' of the vector field $\mathbf{u}$.

Physical interpretation of curl

Again think of $\mathbf{A}$ as a fluid flow. Imagine a small archery arrow whose base (opposite the sharp arrowhead) is fixed to a static point $\mathbf{r}$ of interest, but whilst allowing for free rotation of the arrow. The fluid may rotate the arrow by pushing its feathers.

The arrow rotates fastest when pointing parallel to the curl. Of the two choices, the curl is chosen so that the arrow rotates *anticlockwise* when viewed from above, looking down at the sharp end. This is the *right hand rule*: give a thumbs up with your right hand and your clasped fingers curl anticlockwise, when looking down onto your hand with your thumb (the direction of the curl) pointing upwards.

The speed of rotation corresponds to the length of the curl vector.

So the curl $\nabla \times \mathbf{A}$ determines, at each point, the speed and direction of maximum circulation of $\mathbf{A}$.

5.8 Combinations of grad, div and curl

Grad, div and curl can be combined together in several ways. Suppose ϕ is a scalar field, then

$$(i) \operatorname{div}(\operatorname{grad} \phi) = \nabla \cdot (\nabla \phi) = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = \nabla^2 \phi$$

To show this, let $\mathbf{A} = \nabla \phi$ then

$$\mathbf{A} = (A_1, A_2, A_3) = \left(\frac{\partial \phi}{\partial x}, \frac{\partial \phi}{\partial y}, \frac{\partial \phi}{\partial z} \right)$$

and

$$\begin{aligned} \nabla \cdot (\nabla \phi) = \nabla \cdot \mathbf{A} &= \frac{\partial A_1}{\partial x} + \frac{\partial A_2}{\partial y} + \frac{\partial A_3}{\partial z} \\ &= \frac{\partial}{\partial x} \left(\frac{\partial \phi}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial \phi}{\partial y} \right) + \frac{\partial}{\partial z} \left(\frac{\partial \phi}{\partial z} \right) \\ &= \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2}. \end{aligned}$$

$\nabla^2 \phi$ is called the Laplacian of ϕ , and sometimes written as Δ .

$$(ii) \text{ curl}(\text{grad } \phi) = \nabla \times (\nabla \phi) = \mathbf{0}$$

as

$$\begin{aligned} \nabla \times \nabla \phi &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial \phi}{\partial x} & \frac{\partial \phi}{\partial y} & \frac{\partial \phi}{\partial z} \end{vmatrix} \\ &= \mathbf{i} \left(\frac{\partial^2 \phi}{\partial y \partial z} - \frac{\partial^2 \phi}{\partial z \partial y} \right) - \mathbf{j} \left(\frac{\partial^2 \phi}{\partial x \partial z} - \frac{\partial^2 \phi}{\partial z \partial x} \right) \\ &\quad + \mathbf{k} \left(\frac{\partial^2 \phi}{\partial x \partial y} - \frac{\partial^2 \phi}{\partial y \partial x} \right) \\ &= \mathbf{0} \end{aligned}$$

$$(iii) \text{ div}(\text{curl } \mathbf{A}) = \nabla \cdot (\nabla \times \mathbf{A}) = 0.$$

Aside: Physical interpretation of Laplacian

The Laplacian $\Delta\phi$ from (i) above will appear again in the final chapter.

Intuitively, the Laplacian $\Delta\phi$ is large (and positive) at $\mathbf{r}$ if ϕ tends to be larger than $\phi(\mathbf{r})$ in a small region about $\mathbf{r}$. Conversely, $\Delta\phi$ is negative at $\mathbf{r}$ if, on average, the value of ϕ is smaller in the immediate vicinity of $\mathbf{r}$ than $\phi(\mathbf{r})$.

This is analogous to $\frac{d^2f}{dx^2}$ in the scalar, single-variable case: consider in particular positivity/negativity at troughs/peaks.

$\Delta\phi$ is 0 at $\mathbf{r}$ if $\phi(\mathbf{r})$ agrees with the average of the values of $\phi(\mathbf{r}')$ with $\mathbf{r}'$ in a small region around $\mathbf{r}$. Physically, we would expect this for a temperature distribution which has stabilised: points where $\Delta\phi$ are positive/negative would lose/gain heat from their neighbours.

This is exactly the intuition for the heat equation, which we will see in the final chapter.

5.9 Grad, div and curl of products

Suppose f and g are scalar fields and $\mathbf{A}$ and $\mathbf{B}$ are vector fields. Then the following rules hold.

Rules for ∇

- (i) $\text{grad}(fg) = \nabla(fg) = f\nabla g + g\nabla f,$
- (ii) $\text{div}(f\mathbf{A}) = \nabla \cdot (f\mathbf{A}) = \nabla f \cdot \mathbf{A} + f\nabla \cdot \mathbf{A},$
- (iii) $\text{curl}(f\mathbf{A}) = \nabla \times (f\mathbf{A}) = \nabla f \times \mathbf{A} + f\nabla \times \mathbf{A},$
- (iv) $\text{div}(\mathbf{A} \times \mathbf{B}) = \nabla \cdot (\mathbf{A} \times \mathbf{B}) = (\nabla \times \mathbf{A}) \cdot \mathbf{B} - (\nabla \times \mathbf{B}) \cdot \mathbf{A}.$

5.10 Line integration and Green's Theorem

In this chapter we have defined grad, div and curl of fields in a three dimensional space. While grad and div can be defined completely analogously in any number of dimensions, curl cannot.

However, there is a notion of 2-dimensional curl (note: it is not always called 'curl', so we will denote it by curl_{2d}).

Take a vector field $\mathbf{A}$ in the plane, so $\mathbf{A} = (P, Q)$ for scalar functions $P(x, y)$, $Q(x, y)$. We define:

$$\text{curl}_{2d}\mathbf{A} = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = Q_x - P_y.$$

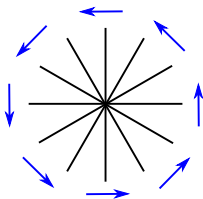
Example 5.10.1

Let $\mathbf{A}(x, y) = (2x^3 - 2y, x \cos(y))$. Then

$$\text{curl}_{2d}\mathbf{A} = \frac{\partial}{\partial x}(x \cos(y)) - \frac{\partial}{\partial y}(2x^3 - 2y) = \cos(y) - 2$$

Physical interpretation of curl_{2d}

Imagine $\mathbf{A}$ as a 2-dimensional fluid flow. Fix a tiny paddle-wheel at the point $\mathbf{r}$ of interest: it may spin about its central axis, located at $\mathbf{r}$. Then $\text{curl}_{2d}\mathbf{A} > 0$ means that the wheel rotates anticlockwise, spinning very quickly if the curl is large. If the curl is negative then rotation will be clockwise.

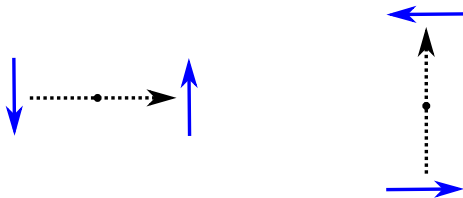


In the image above, the paddle wheel (in black) has centre at $\mathbf{r}$, with the direction of $\mathbf{A}$ indicated in blue nearby: we see that the wheel will rotate anticlockwise, so $\text{curl}_{2d}(\mathbf{A}) > 0$ at $\mathbf{r}$.

The above interpretation makes sense by thinking about the quantity $Q_x - P_y$ for a vector field $\mathbf{A}(x, y) = (P(x, y), Q(x, y))$.

For Q_x , consider the left-hand figure below. As one travels in the x direction (the dotted arrow), in this example the y -component of the vector field (given by Q) goes from negative to positive, so $Q_x > 0$, which pushes the wheel anticlockwise.

Similarly, P_y measures the change in the flow in the x direction as one goes from below $\mathbf{r}$ to above it. This is negative (see right-hand figure), for example, when the flow below the wheel is to the right, and the flow above is to the left (in the $-x$ direction), explaining the sign of $-P_y$. The overall rotation is the net contribution of these orthogonal directions.



Line integration of vector fields

Let $\mathbf{A} = (P, Q)$ be a vector field and C be a 'curve' between two points $\mathbf{u}$ and $\mathbf{v}$ in the plane. Explicitly, we parametrise C as a path $\mathbf{r}(t) = (r_1(t), r_2(t))$ for times t between a and b , so that $\mathbf{r}(a) = \mathbf{u}$ and $\mathbf{r}(b) = \mathbf{v}$.

We define the **line integral** of $\mathbf{A}$ along C to be

$$\int_C \mathbf{A} \cdot d\mathbf{r} = \int_a^b \mathbf{A}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt, \text{ where } \mathbf{r}'(t) = (r_1'(t), r_2'(t)).$$

This does not depend on the choice of parametrisation $\mathbf{r}$ of the curve C between $\mathbf{u}$ and $\mathbf{v}$, but it does depend on the direction along C (if one reverses direction, to travel from $\mathbf{v}$ to $\mathbf{u}$, one reverses the sign of the line integral).

The following notation for the line integration is also often used:

$$\int_C P dx + Q dy.$$

Example 5.10.2

Let $\mathbf{A}(x, y) = (0, -x)$. Find the line integral of $\mathbf{A}$ given by the (anticlockwise) path around the unit circle C centred at 0.

First, we may parametrise the path as $\mathbf{r}(t) = (\cos t, \sin t)$ for t between 0 and 2π , and thus $\mathbf{r}'(t) = (-\sin t, \cos t)$.

By the definition of the line integral,

$$\begin{aligned}\int_C \mathbf{A} \cdot d\mathbf{r} &= \int_0^{2\pi} \mathbf{A}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_0^{2\pi} (0, -\cos t) \cdot (-\sin t, \cos t) dt = \\ &= \int_0^{2\pi} -\cos^2 t = \int_0^{2\pi} \frac{-\cos(2t) - 1}{2} dt = -\pi\end{aligned}$$

Try drawing a sketch of the vector field. Does it make sense that the flow generally works against you? (that is, that the line integral is negative).

Physical interpretation of line integrals

Once again, imagine $\mathbf{A}$ as representing a fluid flow.

The vector $\mathbf{r}'(t)$ gives the direction of travel along the path at time t . If this agrees with $\mathbf{A}$ at the corresponding point $\mathbf{r}(t)$ in the path, then $\mathbf{A}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)$ will be positive. This is the case when $\mathbf{A}$ is pushing in the same direction as the motion of travel. When this dot product is negative, we're working against the flow.

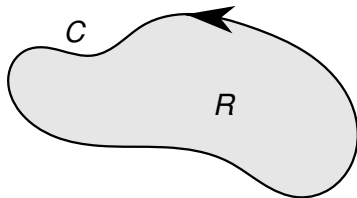
The line integral is the net effect of this over the whole path. It corresponds to the work done by $\mathbf{A}$ in travelling along the path $\mathbf{r}$.

Connection between 2d-curl and line integrals

Suppose C is a simple closed curve in the plane: it does not cross itself and starts where it ends, enclosing a region R of the plane. In this case, the line integral is often denoted

$$\oint_C \mathbf{A} \cdot d\mathbf{r}.$$

to emphasise the circular shape of C . Parametrise C with a path $\mathbf{r}(t)$ with an **anti-clockwise orientation**. We say that C is **positively oriented**. This means R stays on your left as you walk around C :



Connection between 2d-curl and line integrals (ctd)

Take any fixed $\mathbf{r}$ and consider shrinking C to a tiny loop around $\mathbf{r}$. The line integral will of course get very small. However, if we rescale by the area of the region R inside C then,

$$\frac{\oint_C \mathbf{A} \cdot d\mathbf{r}}{\text{area of } R} \approx \text{curl}_{2d} \mathbf{A}(\mathbf{r})$$

with the approximation approaching equality as C shrinks to a point.

This means that $\text{curl}_{2d} \mathbf{A}$ represents the infinitesimal ‘circulation’ per unit area of the vector field. There is an analogous (although slightly more complicated) such interpretation in the 3-dimensional setting.

George Green.

NOTTINGHAM'S MAGNIFICENT MATHEMATICIAN

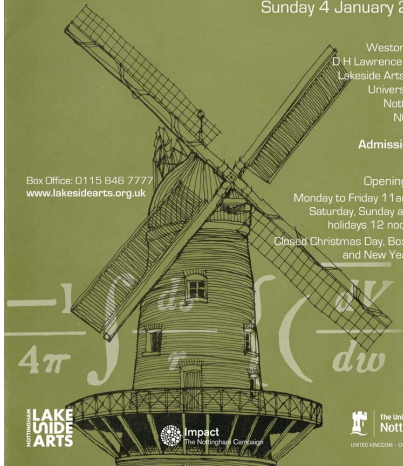
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Green's Theorem

Theorem (Green's Theorem)

For a positively oriented simple closed curve C and vector field $\mathbf{A}$ in the plane,

$$\oint_C \mathbf{A} \cdot d\mathbf{r} = \iint_R \text{curl}_{2d} \mathbf{A}(x, y) \, dx dy.$$

This theorem is more commonly stated as:

$$\oint_C P \, dx + Q \, dy = \iint_R \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \, dx \, dy$$

Example 5.10.3

Consider $\mathbf{A}(x, y) = (0, -x)$ from the previous example. Then

$$\text{curl}_{2d} \mathbf{A} = \frac{\partial}{\partial x}(-x) - \frac{\partial}{\partial y}(0) = -1.$$

So the double integral on the RHS of Green's Theorem is the area of the unit circle, with flip of sign. This agrees with $\oint_C \mathbf{A} \cdot d\mathbf{r} = -\pi$ which we calculated earlier.

Intuitively, Green's theorem says that the macroscopic 'circulation' around the curve C (the tendency for the flow to push with or against the path) is the 'sum' of microscopic circulations (the 2d-curl) inside the curve C .

To see why this makes sense, imagine breaking R into many tiny regions with simple closed curves as boundaries. Adding together all of these line integrals gives the same left-hand side, since the inner edges (not on C) have contributions which cancel with their (oppositely oriented) adjacent neighbours. As we've seen, these very tiny line integrals about closed loops correspond to the 2d-curl, after multiplying by the area inside these curves. On the other hand, summing up the 2d curl about many tiny regions, which we scale by multiplying by the area of each small region, is precisely to take the double integral of the curl over R !

Warning: Scalar line integrals

In this part of the notes we have discussed line integrals of **vector fields** $\mathbf{A}(x, y)$. There is also a notion of the line integral of a **scalar field** $f(x, y)$ along a curve C . It is defined as:

$$\int_C f \, ds = \int_a^b f(\mathbf{r}(t)) |\mathbf{r}'(t)| \, dt$$

where s stands for 'arc length'. In the above, $\mathbf{r}(t)$ is a curve parametrising C for times t between a and b .

The above line integral can be thought of as the (signed) area between the graph of f and the curve C , similarly to in the one-variable case where we integrate over straight line-segments in the x -axis, that is, between points a and b (with little choice for the path).

Reparametrising

The term $|\mathbf{r}'(t)|$ in the above definition is the speed one is travelling along the path at time t . This term ensures that the line integral does not depend on the parametrisation $\mathbf{r}$ chosen for the curve C .

Note that (unlike for the line integral of a vector field) reversing the direction of travel also preserves the line integral in this case: it *does not* reverse the sign, as it does for the line integral of a vector field!

Both the scalar and vector versions of line integrals extend to more variables, using analogous definitions.